

Advanced GIS and Remote Sensing

Toward an Optimized Rainfall Monitoring Network across the Kellerwald Region

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Course objective

Throughout this course you will learn how to **design, analyse, and evaluate a spatially explicit hydrological monitoring network**.

Your overarching goal is to develop a *reproducible and defensible rain-gauge network concept* for the Kellerwald region (~200 km²).

This is not a purely technical mapping task, but a research-oriented design challenge that connects spatial reasoning, remote-sensing data, and hydrological process understanding.

The project evolves in three conceptual phases:

1. **Exploration** – Identify what makes a precipitation network “good” or “bad”, and define what “optimal” could mean in this context.
2. **Development** – Apply methods learned in class (Kriging, radar bias correction, RS integration, uncertainty analysis) to iteratively refine your network.
3. **Synthesis** – Present and defend a transparent, reproducible network design that links rainfall monitoring to hydrological response at sub-catchment scale.

Final deliverables

At the end of the course each group submits **three complementary products**:

1. Transparent repository (core submission)

A public or shared repository (e.g. GitHub, GitLab, or institutional server) that contains: - All data sources, scripts, and documentation used to derive your final network.

- A reproducible workflow that shows how you went from raw data to station selection and evaluation.

- Clear version control and short README explaining structure and logic.

The repository represents the **traceable backbone** of your project and demonstrates reproducibility and collaboration.

2. Poster

A concise visual summary of: - The final proposed network (map and rationale).

- The main methodological steps (e.g. interpolation, uncertainty analysis).

- Key results or validation indicators.

- A short reflection on limitations and future improvement.

The poster must communicate the project as if presented at a professional hydrology or GIS conference.

3. Five-minute pitch (presentation)

Each group presents their poster in a short oral pitch (max 5 min).

The goal is to **argue convincingly** why your design is optimal *for its chosen criteria* (coverage, efficiency, hydrological adequacy).

The pitch should demonstrate technical understanding, critical reasoning, and teamwork.

Evaluation criteria (final synthesis)

Each submission is evaluated across **five dimensions**, 0–2 points each (max 10 pts):

Criterion	0	1	2
Spatial reasoning	arbitrary placement	partial coverage of key gradients	systematic physiographic logic (Luv–Lee, elevation, forest density)
Hydrological integration	no link to runoff	implicit link to subcatchments	explicit precipitation–discharge coupling and closure logic
Methodological rigour	ad-hoc reasoning	partial quantitative justification	reproducible workflow using kriging, RS, or uncertainty metrics
Transparency & reproducibility	no documentation	partial scripts / missing metadata	complete, documented, and reproducible repository
Communication quality	unclear poster / pitch	partially coherent argument	clear, concise, and scientifically convincing presentation

Competition element

The Kellerwald Network Design Project is structured as a **friendly competition**: the group achieving the highest total score becomes the *Course Design Champion*.

In case of a tie, the tiebreaker is **Hydrological integration**, since the final goal is a coherent rainfall–runoff framework, not aesthetic map design.

Learning outcome

By completing the full project you will be able to: - Formulate and test design hypotheses for spatial monitoring networks.

- Apply geostatistical, RS, and modelling tools to evaluate network performance.
- Document and communicate your workflow transparently.
- Defend a reasoned, evidence-based concept of “optimality” in hydrological monitoring.